

Setting & Problem Formulation

Classical Constrained RL

Finite discounted MDP, unknown dynamics $\mathbb{P}$, initial distribution μ_0 , and $\gamma \in (0, 1)$.

Reward under a cost budget

$$\max_{\pi \in \Pi} J_r(\pi) \quad \text{s.t.} \quad J_c(\pi) \leq b.$$

For reward or cost $q \in \{r, c\}$,

$$J_q(\pi) = (1 - \gamma) \mathbb{E}_{\mu_0, \pi} \left[\sum_{t=0}^{\infty} \gamma^t q(s_t, a_t) \right].$$

$$= \sum_{s,a} (1 - \gamma) \underbrace{\sum_{t=0}^{\infty} \gamma^t \mathbb{P}_{\mu_0, \pi}(s_t = s, a_t = a)}_{=: \lambda^\pi(s, a)} \cdot q(s, a)$$

State-Action Occupancy Measure

Represent a policy by its **discounted state-action visitation**:

$$\lambda^\pi(s, a) = (1 - \gamma) \sum_{t=0}^{\infty} \gamma^t \cdot \mathbb{P}_{\mu_0, \pi}(s_t = s, a_t = a).$$

$$J_r(\pi) = \langle r, \lambda^\pi \rangle, \quad J_c(\pi) = \langle c, \lambda^\pi \rangle.$$

Hidden Convex Transformation [1,2]

With softmax policy $\pi_\theta(a|s) := \frac{\exp(\psi(s, a; \theta))}{\sum_{a' \in \mathcal{A}} \exp(\psi(s, a'; \theta))}$, the mapping

$$\theta \mapsto \lambda^{\pi_\theta}$$

is locally invertible, Lipschitz inverse and *locally hidden convex*, i.e. $(1 - \alpha)\lambda^{\pi_\theta} + \alpha\lambda^* \in \mathcal{V}_\theta$

Constrained Maximum Exploration

In absence of a reward function, explore maximally:

$$\min_{\theta \in \Theta} -\mathcal{H}(\lambda^{\pi_\theta}) := \sum_{s,a} \lambda^{\pi_\theta}(s, a) \log \lambda^{\pi_\theta}(s, a) \\ \text{s.t.} \quad \mathcal{R}(\lambda^{\pi_\theta}) \leq 0.$$

Constraints beyond linear structure:

- Cost budget:

$$\mathcal{R}(\lambda) := \langle c, \lambda \rangle - c_{\max}$$

- Apprenticeship Learning:

$$\mathcal{R}(\lambda) := D_{\text{KL}}(\lambda \| \lambda^{\text{ref}}) - \mathcal{R}_{\max}.$$

$-\mathcal{H}, \mathcal{R}$ Convex in λ , generally non-convex in policy parameters $\rightarrow$ **no add. Bellman structure**

Pseudo-Rewards [3]:

Differentiable F , $\lambda(\theta) := \lambda^{\pi_\theta}$ and $r_F := \nabla_\lambda F(\lambda(\theta))$, Policy Gradient Thm.: $\nabla_\theta F(\lambda(\theta)) = [\nabla_\theta \lambda(\theta)]^\top r_F = \nabla_\theta V^{\pi_\theta}(r) \Big|_{r=\nabla_\lambda F(\lambda(\theta))}$.

Approach & Analysis

Policy Gradient Penalty Method

Replace the constraint by a quadratic penalty:

$$P_\beta(\theta) := -\mathcal{H}(\lambda(\theta)) + \beta[\mathcal{R}(\lambda(\theta))]_+^2$$

1. Sample and estimate occupancy

At iteration k , sample B trajectories of length H under π_{θ_k} :

$$\hat{\lambda}^{(k)} = \frac{1 - \gamma}{B} \sum_{b=1}^B \sum_{t=0}^{H-1} \gamma^t \delta_{s_t^{(b)}, a_t^{(b)}}.$$

2. Construct pseudo-rewards via chain rule

$$g_{\mathcal{H}} := \nabla_\lambda(-\mathcal{H}(\lambda)) \Big|_{\lambda=\hat{\lambda}^{(k)}}, \quad g_{\mathcal{R}} := \nabla_\lambda([\cdot]_+^2 \circ \mathcal{R}(\lambda)) \Big|_{\lambda=\hat{\lambda}^{(k)}}$$

3. Reuse the batch for REINFORCE

$$\hat{g}_k = \frac{1}{B} \sum_{b=1}^B \text{REINFORCE}(\tau_b; g_{\mathcal{H}} + \beta g_{\mathcal{R}}).$$

4. Update the policy parameters

$$\theta_{k+1} = \text{proj}_\Theta[\theta_k - \eta \hat{g}_k].$$

Main Contribution

- Single-loop, primal-only** method, operating directly in the policy space with single-batch penalty gradient estimation

- Global, last-iterate convergence** under biased stochastic gradient; only strong duality needed, no Slater's condition

- Late-iterate stability and scalability** to continuous control tasks beyond finite grid worlds covered by theory

Assumptions

Regularity. L-smooth softmax parametrization.

Hidden convexity. $\theta \mapsto \lambda^{\pi_\theta}$ locally hidden convex, inverse Lipschitz and locally invertible.

Strong duality. Finite Lagrange multiplier $\nu^* \geq 0$.

Convergence Theorem

For target accuracy $\varepsilon > 0$, choose

$$\eta = \mathcal{O}(\varepsilon), \quad \beta = \mathcal{O}(\varepsilon^{-1}), \\ B = \mathcal{O}(\varepsilon^{-4}), \quad H = \mathcal{O}(\log(1/\varepsilon)).$$

After $N = \tilde{\mathcal{O}}(\varepsilon^{-2})$ iterations, the **last policy** satisfies

$$-\varepsilon \leq \mathbb{E}[\mathcal{H}(\lambda^{\pi_{\theta(N)}}) - \mathcal{H}^*] \leq \varepsilon, \quad \mathbb{E}[[\mathcal{R}(\lambda^{\pi_{\theta(N)}})]_+] \leq \varepsilon.$$

Here $\mathcal{H}^*$ is the optimal feasible entropy.

Total complexity: $\tilde{\mathcal{O}}(\varepsilon^{-6})$

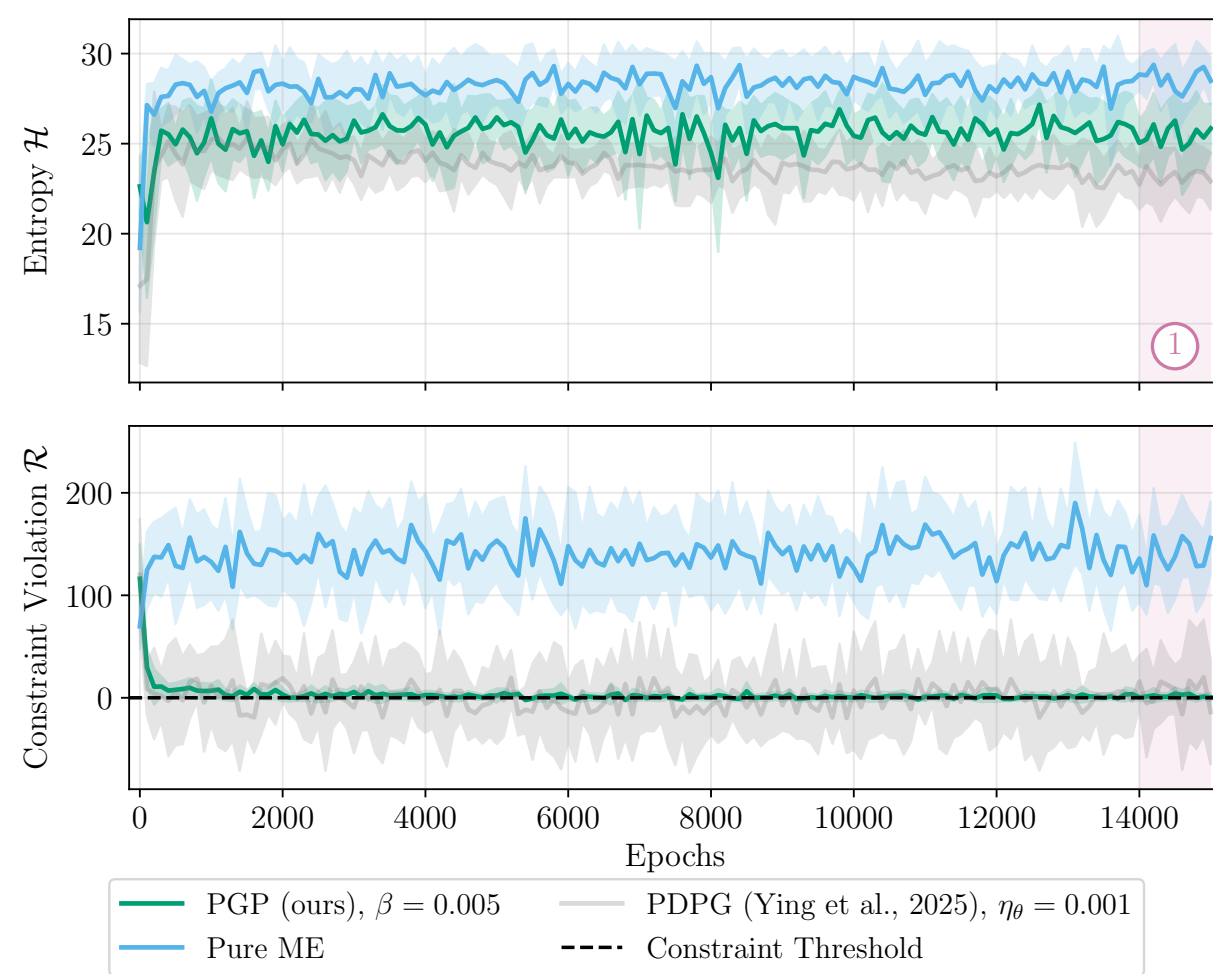
Link to Paper:



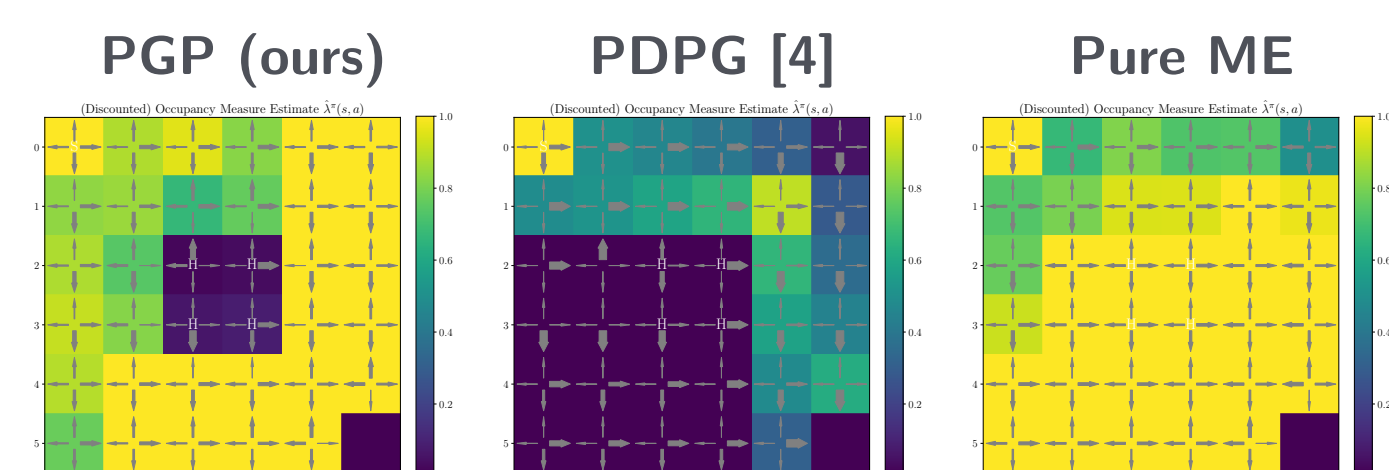
arXiv:2604.28144

Experimental Results

Theory: Gridworld



Mean $\pm$ standard deviation over 10 seeds.



Last-Iterate Stability

Fraction of late-training checkpoints meeting each criterion.

① Gridworld

	$\mathcal{R} \leq 0.1$	$\mathcal{R} \leq 0.01$
PGP (ours)	97.21%	92.43%
PDPG	65.33%	51.01%

Final 1,000 epochs over 10 seeds: 10,000 checkpoints.

② SafeCartpole

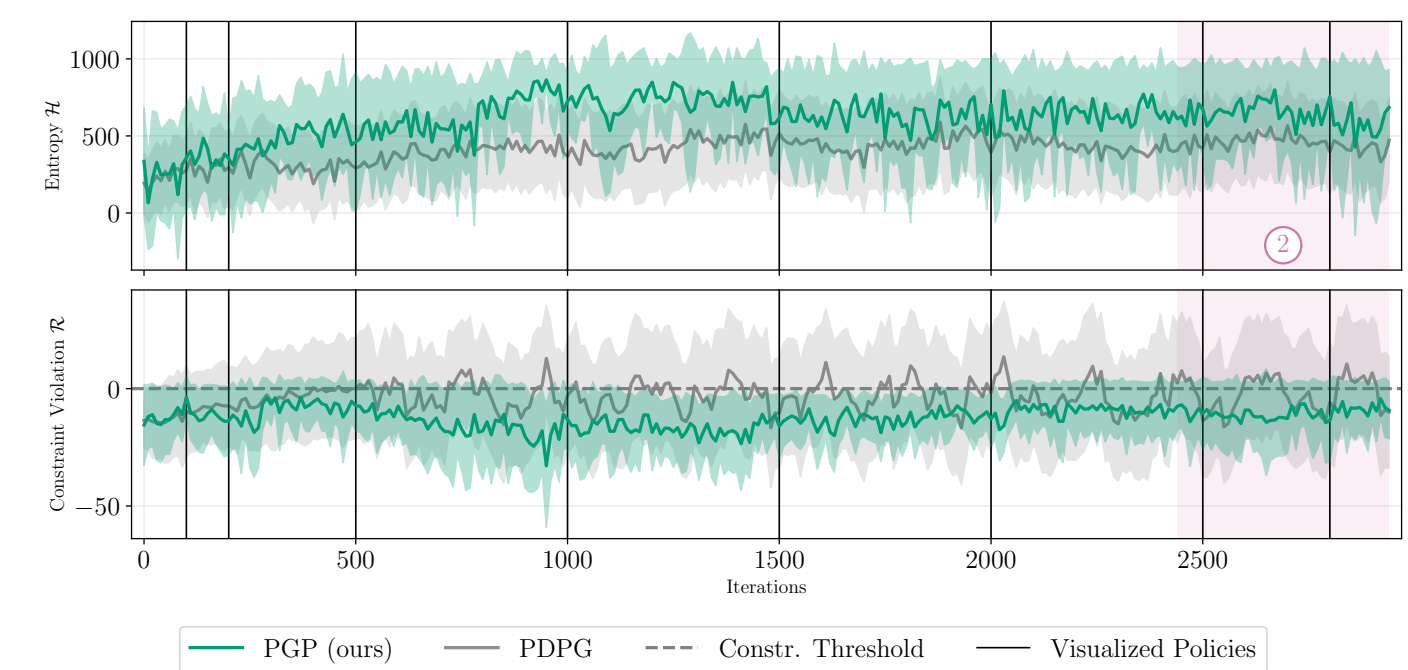
	PGP (ours)	PDPG
$T_{\text{up}} > 1$	87.23%	59.47%
$T_{\text{up}} > 60$	86.11%	41.64%
$\mathcal{R} \leq 0.1$	96.89%	70.03%
$\mathcal{R} \leq 0.01$	94.49%	53.33%

T_{up} : sustained swing-up steps.

Final 500 epochs over 5 seeds: 2,500 checkpoints.

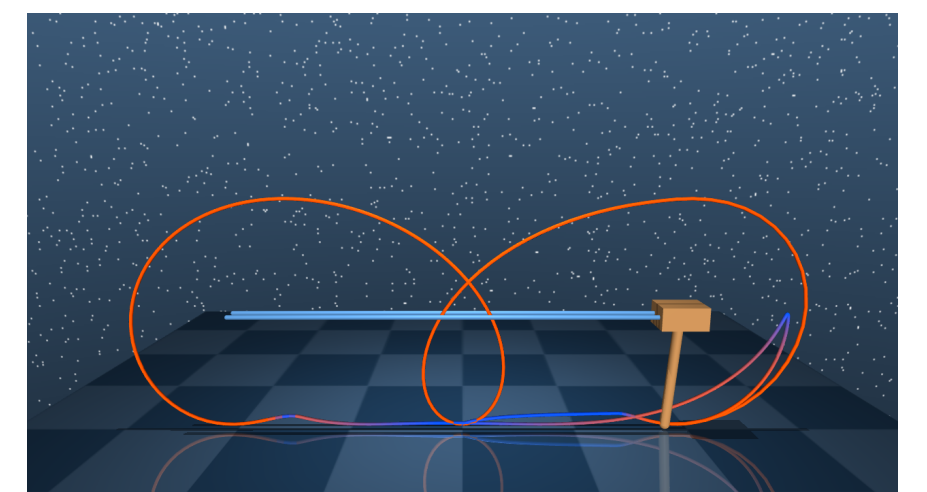
Beyond Theory: Scaling SafeCartpole

Continuous $\mathcal{S}, \mathcal{A}$; $\hat{\lambda}$ estimated with Gaussian mixture model



Mean $\pm$ standard deviation over 5 seeds.

PGP Learned Trajectory Sample, $t = 9.6$ s



Swing-up and symmetrical exploration near the slider limits, **tracer color** denotes pole **angular velocity** (lower, higher).